

Aaron Leevord Fernandes

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EDUCATION

Indiana University Bloomington

Aug 2023 – May 2025

M.S. in Intelligent Systems Engineering (Computer Engineering)

GPA: 3.9/4.0

Goa University

Aug 2017 – Jul 2021

B.E. in Electronics and Telecommunication

GPA: 3.7/4.0

EXPERIENCE

Research Data Scientist (Computer Vision)— Leung Research Group, Indiana University

Apr 2024 – Present

- Designed an unpaired OCT-video → histology translation pipeline (CNN-GRU encoder-decoder + Transformer), reducing FID by 50% vs a Fourier-domain baseline and improving visual realism of generated tissue morphology.
- Implemented diffeomorphic registration (VoxelMorph-style U-Net + spatial transformer), reducing frame-to-frame alignment error by 10% (reconstruction MSE).
- Architected distributed FDTD simulation pipeline on HPC cluster with Slurm and MPI parallelization, generating 5K+ synthetic samples with 90% wall-clock time reduction; developed U-Net surrogate models achieving 70% speedup over Maxwell solvers at 95% accuracy

Machine Learning Engineer (Reinforcement Learning) – Indiana University

Jan 2025 – June 2025

- [GitHub: multi-goal-rl-svgg](#)
- Led development of a multi-goal, sparse-reward RL pipeline in MuJoCo, designing custom benchmarks (FetchReach, PointMaze w/ obstacles) and evaluation tooling for goal-conditioned policies.
- Implemented goal-conditioned SAC + HER and improved success rate from 15% to 55% through replay optimization and goal relabeling strategies.
- Implemented SVGG curriculum learning (KDE-driven goal selection) and scaled experimentation to 100+ tracked runs with Hydra + W&B, enabling reproducible sweeps and clean ablations.

Software Engineer — Tata Consultancy Services (HDFC Bank)

Jan 2022 – Jan 2023

- Architected microservices-based transaction processing system in Java using Spring Boot, Hibernate ORM, and RESTful APIs, handling 100K+ daily transactions with 99.9% uptime and sub-200ms p99 latency
- Implemented automated reconciliation workflows decreasing error rate from 12% to 3%

SELECTED PROJECTS

ComicPaliGemma – Visual Language Model — PyTorch, CUDA, Triton

July 2024 – Present

- Built a from-scratch implementation of Google's PaliGemma VLM (SigLIP ViT vision encoder + Gemma decoder), validating full pretrained checkpoint compatibility.
- Fine-tuned on a 5K+ comic-panel dataset using **QLoRA (4-bit)** to reduce peak GPU VRAM by **65%** while maintaining model quality.
- Developed a long-context autoregressive text generator conditioned on frozen VLM embeddings for **coherent narrative generation** across 10+ comic panels; integrated **Triton/FlashAttention-2** style attention kernels to achieve 5–10× faster inference vs standard PyTorch attention.

Diffusion Models (PyTorch)

Dec 2025 – Ongoing

- [Github: Building Diffusion Models](#)
- Reproduced a DDPM baseline from first principles in PyTorch (U-Net, timestep conditioning, attention, ancestral sampling), validating core diffusion equations and training dynamics end-to-end.
- Designed the project for controlled research iteration: modular noise schedules, objectives, and samplers with reproducible ablations to study modern variants (DDIM, score-based methods, consistency models).

Model-Based RL with Model Predictive Control — PyTorch, Mujoco

Aug 2024 – Dec 2024

- [Github: Model Based RL with MPC](#)
- Implemented a learned world model based algorithm, planning with Cross-Entropy Method (CEM) and Model Predictive Control (MPC) for trajectory optimization; achieved 2× sample efficiency vs SAC baseline on HalfCheetah and Reacher

Predictive Modeling for NCAA Women's Basketball (Random Forests)

Sep 2024 - Oct 2024

- Built **predictive Random Forest models** on large-scale customer behavior data, improving accuracy by **19%** and ranking **top 5/67** teams in national competition. And delivered business insights that projected up to **15% revenue growth** and **10% fan engagement gains**.

TECHNICAL SKILLS

ML Specializations: Computer Vision, Diffusion Models, Image Processing, RL, Bayesian Inference, NLP (LLMs, Finetuning, RLHF and RAG)

ML Frameworks & Libraries: PyTorch (incl. torch.profiler), TensorFlow, JAX, Triton, Ray, vLLM, Gymnasium (Mujoco), HuggingFace Transformers, Scikit-learn, NumPy, Pandas, Matplotlib, MLFlow, Weights & Biases

Cloud/HPC: AWS (SageMaker, EC2, S3, Lambda), Docker, Kubernetes, Slurm, MPI, Spark, Hadoop

Programming: Python (expert), C++ (CUDA/system programming), Java (Spring Boot), JavaScript/TypeScript, SQL, PostgreSQL, MongoDB, Neo4j, Redis, GraphQL, REST APIs

Certifications: [AWS Developer Associate](#), [AWS Cloud Practitioner](#)

Advanced Coursework & Self-Study: DeepBayes Summer School (Bayesian Deep Learning), Sutton & Barto RL, Bishop's Pattern Recognition & Machine Learning, Diffusion Models (DDPM, DDIM, Score-based Generative Models papers)